Solutions

Maths

Q1. If $f(x) = \tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right) + \cos^{-1}(2x^2-1)$ for $x \in \left(0, \frac{1}{\sqrt{2}}\right)$, then $f(x)$ is equal to:

(A) $\frac{\pi}{2} + \frac{3}{2}\sin^{-1}x$ (B) $\pi - \frac{1}{2}\sin^{-1}x$ (C) $\pi - \frac{3}{2}\sin^{-1}x$ (D) $\frac{1}{2}\sin^{-1}x$

Ans. C

Sol. Let $x = \sin\theta$. Since $x \in \left(0, \frac{1}{\sqrt{2}}\right)$, we have $\theta \in \left(0, \frac{\pi}{4}\right)$.

$$\tan^{-1}\left(\frac{\sin\theta}{1+\cos\theta}\right) = \tan^{-1}\left(\frac{2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)}{2\cos^2\left(\frac{\theta}{2}\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right)$$

Since $\theta \in \left(0, \frac{\pi}{4}\right) \implies \frac{\theta}{2} \in \left(0, \frac{\pi}{8}\right)$, $\tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right) = \frac{\theta}{2} = \frac{1}{2}\sin^{-1}x$

$$\cos^{-1}(2x^2-1) = \cos^{-1}(2\sin^2\theta - 1)$$

$$= \cos^{-1}(-\cos 2\theta) = \pi - \cos^{-1}(\cos 2\theta)$$

Since $\theta \in \left(0, \frac{\pi}{4}\right) \implies 2\theta \in \left(0, \frac{\pi}{2}\right)$, $\cos^{-1}(\cos 2\theta) = 2\theta = 2\sin^{-1}x$

So, $\cos^{-1}(2x^2-1) = \pi - 2\sin^{-1}x$.

$$f(x) = \frac{1}{2}\sin^{-1}x + \pi - 2\sin^{-1}x = \pi - \frac{3}{2}\sin^{-1}x$$

Q2. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function satisfying $f(x+y) = f(x) + f(y) + 2xy$ for all $x, y \in \mathbb{R}$. If $\lim\limits_{x\to 0}\frac{f(x)}{x} = 1$, then the function $g(x) = |f(x)|$ is non-differentiable at:

(A) exactly one point (B) exactly two points (C) exactly three points (D) no point in $\mathbb{R}$

Ans. B

Sol. By the definition of derivative: $f'(x) = \lim\limits_{h\to 0}\frac{f(x+h) - f(x)}{h}$

Using the given functional relation $f(x+h) = f(x) + f(h) + 2xh$:

$$f'(x) = \lim_{h\to 0}\frac{f(h) + 2xh}{h} = \lim_{h\to 0}\frac{f(h)}{h} + 2x$$

Given $\lim\limits_{h\to 0}\frac{f(h)}{h} = 1$, $f'(x) = 2x + 1$

Integrating both sides with respect to x: $f(x) = x^2 + x + C$

Putting $x = y = 0$ in the functional equation yields $f(0) = 0$, so $C = 0$. Thus: $f(x) = x^2 + x = x(x+1)$

Now consider $g(x) = |f(x)| = |x(x+1)|$.

The critical points where $f(x) = 0$ are $x = 0$ and $x = -1$.

At $x = 0$, $f'(0) = 1 \neq 0$. At $x = -1$, $f'(-1) = -1 \neq 0$.

Since $f'(x) \neq 0$ at the simple roots of $f(x)$, the function $g(x) = |f(x)|$ is non-differentiable at $x = 0$ and $x = -1$.

Thus, $g(x)$ is non-differentiable at exactly two points.

Q3. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function defined by $f(x) = \begin{cases} x^3 + 2x + a, & \text{if } x \geq 0 \\ cx - 2, & \text{if } x < 0 \end{cases}$ where a and c are real constants. If $f^{-1}(x)$ exists for all $x \in \mathbb{R}$, then the sum of all real values of x satisfying the equation $f(x) = f^{-1}(x)$ is equal to:

(A) 0 (B) 1 (C) 2 (D) -1

Ans. B

Sol. Since $f(x)$ is a differentiable function on $\mathbb{R}$, it must be continuous and differentiable at $x = 0$.

For continuity at $x = 0$: $\lim\limits_{x \to 0^+} f(x) = \lim\limits_{x \to 0^-} f(x) = f(0)$ $a = -2$

For differentiability at $x = 0$: $f'(0^+) = f'(0^-)$ $\left.\frac{d}{dx}(x^3 + 2x - 2)\right|_{x=0} = \left.\frac{d}{dx}(cx - 2)\right|_{x=0}$

$3(0)^2 + 2 = c \implies c = 2$

Thus, the function is: $f(x) = \begin{cases} x^3 + 2x - 2, & x \geq 0 \\ 2x - 2, & x < 0 \end{cases}$ Checking the derivative of $f(x)$:

For $x > 0$, $f'(x) = 3x^2 + 2 > 0$. For $x < 0$, $f'(x) = 2 > 0$.

Since $f'(x) > 0$ for all $x \in \mathbb{R}$, $f(x)$ is strictly increasing on $\mathbb{R}$.

For any strictly increasing invertible function, the solutions to $f(x) = f^{-1}(x)$ are identical to the solutions of $f(x) = x$.

Case 1: For $x \geq 0$: $x^3 + 2x - 2 = x \implies x^3 + x - 2 = 0 \implies (x-1)(x^2 + x + 2) = 0$

Since $x^2 + x + 2 = 0$ has no real roots (Discriminant = $1 - 8 = -7 < 0$), the only real root is x = 1, which satisfies x ≥ 0.

Case 2 : For $x < 0$: $2x - 2 = x \implies x = 2$, Since $x = 2 \not< 0$, this solution is rejected.

Q4. Let $A = \{x \in \mathbb{Z} : |x^2 - 4| \leq 12\}$ and $B = \{x \in \mathbb{Z} : x^2 - 4x - 5 \leq 0\}$ be two sets. If a relation R from A to B is defined as $R = \{(a, b) \in A \times B : a + b \text{ is divisible by } 2\}$, then the total number of elements in the relation R is:

(A) 28 (B) 32 (C) 36 (D) 31

Ans. D

Sol. For set A: $|x^2 - 4| \leq 12 \implies -12 \leq x^2 - 4 \leq 12$

$\implies -8 \leq x^2 \leq 16$

Since $x \in \mathbb{Z}$, $x^2 \in \{0, 1, 4, 9, 16\}$, which gives:

$A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ Number of even elements in $A = 5$ (which are

$\{-4, -2, 0, 2, 4\}$)

Number of odd elements in $A = 4$ (which are $\{-3, -1, 1, 3\}$)

For set B: $x^2 - 4x - 5 \leq 0 \implies (x-5)(x+1) \leq 0 \implies -1 \leq x \leq 5$ Since $x \in \mathbb{Z}$:

$B = \{-1, 0, 1, 2, 3, 4, 5\}$

Number of even elements in $B = 3$ (which are $\{0, 2, 4\}$)

Number of odd elements in $B = 4$ (which are $\{-1, 1, 3, 5\}$)

For $(a, b) \in R$, $a + b$ must be even, which occurs when both a and b are even or both a and b are odd.

Number of elements in $R = (\text{Even in } A) \times (\text{Even in } B) + (\text{Odd in } A) \times (\text{Odd in } B)$

$n(R) = (5 \times 3) + (4 \times 4) = 15 + 16 = 31$

Q5. Let $f : \mathbb{R} \to \mathbb{R}$ be a non-uniformly defined function given by $f(x) = \begin{cases} \frac{x^3+\alpha x^2+1}{x^2+2x} - x, & \text{if } [x] \text{ is even} \\ \frac{\beta x^2+3x+2}{2x-1} - \frac{x}{2}, & \text{if } [x] \text{ is odd} \end{cases}$ where $[\cdot]$ denotes the greatest integer function. If $\lim\limits_{x\to\infty} f(x) = L$ exists as a finite real number, then the value of $(4\alpha + 2\beta + 4L)$ is equal to:

(A) 16 (B) 24 (C) 18 (D) 26

Ans. B

Sol. For $\lim\limits_{x\to\infty} f(x) = L$ to exist, the limits of both branches of $f(x)$ as $x \to \infty$ must be equal to L.

Case 1: When $\lfloor x \rfloor$ is even:

$$g_1(x) = \frac{x^3 + \alpha x^2 + 1}{x^2 + 2x} - x = \frac{(\alpha - 2)x^2 + 1}{x^2 + 2x}$$

Taking the limit as $x \to \infty$:

$$\lim_{x\to\infty} g_1(x) = \lim_{x\to\infty} \frac{(\alpha - 2) + \frac{1}{x^2}}{1 + \frac{2}{x}} = \alpha - 2$$

Thus, $L = \alpha - 2$.

Case 2: When $\lfloor x \rfloor$ is odd:

$$g_2(x) = \frac{\beta x^2 + 3x + 2}{2x - 1} - \frac{x}{2} = \frac{2(\beta - 1)x^2 + 7x + 4}{2(2x - 1)}$$

For $\lim\limits_{x\to\infty} g_2(x)$ to be a finite real number, the dominant term coefficient of x^2 in the numerator must be zero.

$2(\beta - 1) = 0 \implies \beta = 1$

Then, $\lim\limits_{x\to\infty} g_2(x) = \lim\limits_{x\to\infty} \dfrac{7x + 4}{4x - 2} = \dfrac{7}{4}$

Thus, $L = \dfrac{7}{4}$.

Equating the limit values from both branches:

$\alpha - 2 = \dfrac{7}{4} \implies \alpha = \dfrac{15}{4}$ Now, evaluating the required expression:

$4\alpha + 2\beta + 4L = 4\left(\dfrac{15}{4}\right) + 2(1) + 4\left(\dfrac{7}{4}\right)$

$= 15 + 2 + 7 = 24$

Hence, the correct option is (B).

Q6. Let $f : \mathbb{R} \to [0,1]$ be a function defined by $f(x) = \sin^2(3x) + \cos^4(3x)$ Then $f(x)$ is:

(A) an onto function with fundamental period $\frac{\pi}{3}$

(B) an into function with fundamental period $\frac{\pi}{6}$

(C) an onto function with fundamental period $\frac{\pi}{6}$

(D) an into function with fundamental period $\frac{\pi}{3}$

Ans. B

Sol. Simplifying the function $f(x)$:

$f(x) = \sin^2(3x) + \cos^2(3x)(1 - \sin^2(3x))$

$f(x) = (\sin^2(3x) + \cos^2(3x)) - \sin^2(3x)\cos^2(3x)$

$f(x) = 1 - \frac{1}{4}\sin^2(6x) = 1 - \frac{1}{4}\left(\frac{1 - \cos(12x)}{2}\right)$

$= \frac{7}{8} + \frac{1}{8}\cos(12x)$

1. Fundamental Period:

The fundamental period of $\cos(12x)$ is $T = \frac{2\pi}{12} = \frac{\pi}{6}$.

Thus, the fundamental period of $f(x)$ is $\frac{\pi}{6}$.

2. Onto vs Into:

Since $-1 \le \cos(12x) \le 1$, we have:

$\frac{7}{8} - \frac{1}{8} \le f(x) \le \frac{7}{8} + \frac{1}{8} \implies$ Range of $f = \left[\frac{3}{4}, 1\right]$

Since Range $\left[\frac{3}{4}, 1\right] \neq$ Codomain $[0,1]$, the function is into.

Q7. Let D be the maximal real domain of the function

$f(x) = \cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right) + \tan^{-1}\left(\frac{2x}{1 - x^2}\right)$

Then the domain D and the simplified value of $f(x)$ for $x \in D \cap (-1, 0)$ are respectively:

(A) $D = \mathbb{R} \setminus \{-1, 1\}$ and $f(x) = 0$ (B) $D = \mathbb{R}$ and $f(x) = 4\tan^{-1}(x)$

(C) $D = \mathbb{R} \setminus \{-1, 1\}$ and $f(x) = -4\tan^{-1}(x)$ (D) $D = (-1, 1)$ and $f(x) = 0$

Ans. A

Sol. For $\cos^{-1}\left(\frac{1 - x^2}{1 + x^2}\right)$ to be defined: $-1 \le \frac{1 - x^2}{1 + x^2} \le 1$

Since $1 + x^2 > 0$ for all $x \in \mathbb{R}$, this inequality holds for all $x \in \mathbb{R}$.

For $\tan^{-1}\left(\frac{2x}{1 - x^2}\right)$ to be defined: The denominator must not be zero:

$1 - x^2 \neq 0 \implies x \neq \pm 1$

Thus, the maximal domain is $D = \mathbb{R} \setminus \{-1, 1\}$.

Let $x = \tan\theta$. For $x \in (-1, 0)$, we have $\theta \in \left(-\frac{\pi}{4}, 0\right)$, which implies $2\theta \in \left(-\frac{\pi}{2}, 0\right)$.

$\cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) = \cos^{-1}(\cos 2\theta)$

Since $2\theta \in \left(-\frac{\pi}{2}, 0\right)$, we have $-2\theta \in \left(0, \frac{\pi}{2}\right)$.

Thus, $\cos^{-1}(\cos 2\theta) = -2\theta = -2\tan^{-1}(x)$

$\tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right) = \tan^{-1}(\tan 2\theta)$

Since $2\theta \in \left(-\frac{\pi}{2}, 0\right) \subset \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, we have: $\tan^{-1}(\tan 2\theta) = 2\theta = 2\tan^{-1}(x)$

Combining both terms for $x \in D \cap (-1, 0)$: $f(x) = -2\tan^{-1}(x) + 2\tan^{-1}(x) = 0$

Q8. In a group of 100 students, three sports—Cricket, Football, and Basketball—are offered. Let A, B, and C be the sets of students playing Cricket, Football, and Basketball respectively, such that $n(A) = 50$, $n(B) = 40$, $n(C) = 35$, $n(A \cap B) = 20$, $n(B \cap C) = 15$, and $n(C \cap A) = 10$.

Let S be the set of all possible cardinalities $\mathrm{n(A \cup B \cup C)}$, and consider the set $P = \{x \in \mathbb{N} : x^2 - 170x + 7200 \leq 0\}$.

Which of the following statements is **CORRECT** ?

(A) S is an infinite set (B) S and P are equal sets (C) S is an empty set

(D) S and P are equivalent but not equal sets

Ans. B

Sol. Let $k = n(A \cap B \cap C)$.

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
$n(A \cup B \cup C) = 50 + 40 + 35 - 20 - 15 - 10 + k = 80 + k$

For the regions in the Venn diagram to be valid (non-negative integer cardinalities):
- $n(A \text{ only}) = 50 - (20 - k + 10 - k + k)$

$= 20 + k \geq 0 \implies k \geq -20$
- $n(B \text{ only}) = 40 - (20 - k + 15 - k + k)$

$= 5 + k \geq 0 \implies k \geq -5$
- $n(C \text{ only}) = 35 - (10 - k + 15 - k + k)$

$= 10 + k \geq 0 \implies k \geq -10$
- $n(A \cap B \text{ only}) = 20 - k \geq 0 \implies k \leq 20$
- $n(B \cap C \text{ only}) = 15 - k \geq 0 \implies k \leq 15$
- $n(C \cap A \text{ only}) = 10 - k \geq 0 \implies k \leq 10$
- $k = n(A \cap B \cap C) \geq 0$

Combining these inequalities gives:$0 \leq k \leq 10$, where $k \in \mathbb{Z}$

Also, $n(A \cup B \cup C) = 80 + k \leq 100 \implies k \leq 20$, which is already satisfied. Thus, $k \in \{0, 1, 2, \ldots, 10\}$.
Therefore, $n(A \cup B \cup C) \in \{80, 81, 82, \ldots, 90\}$.

$S = \{80, 81, 82, \ldots, 90\}$

Now, for set P: $x^2 - 170x + 7200 \leq 0 \implies (x - 80)(x - 90) \leq 0$

$\implies 80 \leq x \leq 90$

Since $x \in \mathbb{N}$: $P = \{80, 81, 82, \ldots, 90\}$

Since S and P contain precisely the same elements, S and P are equal sets ($S = P$).

Q9. Let $f : \left(\frac{1}{\sqrt{3}}, 1\right) \to \mathbb{R}$ be defined by $f(x) = 2\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right) + \sin^{-1}\left(\frac{2x}{1+x^2}\right) - \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ and $g : (0,1) \to \mathbb{R}$ be defined by $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) + 2\tan^{-1}\left(\frac{1-x}{1+x}\right)$

Then for any $x \in \left(\frac{1}{\sqrt{3}}, 1\right)$, the value of $\tan\left(\frac{f(x)+g(x)}{6}\right)$ is equal to:

(A) 2 (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) 1

Ans. D

Sol. For $f(x)$, substitute $x = \tan\theta$. Since $x \in \left(\frac{1}{\sqrt{3}}, 1\right)$, we have $\theta \in \left(\frac{\pi}{6}, \frac{\pi}{4}\right)$.

1. $2\tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right) = 2\tan^{-1}\left(\tan\frac{\theta}{2}\right) = 2\left(\frac{\theta}{2}\right) = \theta$, as $\frac{\theta}{2} \in \left(\frac{\pi}{12}, \frac{\pi}{8}\right)$.
2. $\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right) = \sin^{-1}(\sin 2\theta) = 2\theta$, as $2\theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right) \subset \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
3. $\tan^{-1}\left(\frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}\right) = \tan^{-1}(\tan 3\theta) = 3\theta - \pi$, as $3\theta \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$.

Hence,
$f(x) = \theta + 2\theta - (3\theta - \pi) = \pi$

For $g(x)$, substitute $x = \tan\phi$.

For $x \in (0,1)$, $\phi \in \left(0, \frac{\pi}{4}\right)$.

1. $\cos^{-1}\left(\frac{1-\tan^2\phi}{1+\tan^2\phi}\right) = \cos^{-1}(\cos 2\phi) = 2\phi$, as $2\phi \in \left(0, \frac{\pi}{2}\right)$.
2. $2\tan^{-1}\left(\frac{1-\tan\phi}{1+\tan\phi}\right) = 2\tan^{-1}\left(\tan\left(\frac{\pi}{4} - \phi\right)\right)$

$= 2\left(\frac{\pi}{4} - \phi\right) = \frac{\pi}{2} - 2\phi$.

Hence,
$g(x) = 2\phi + \left(\frac{\pi}{2} - 2\phi\right) = \frac{\pi}{2}$

Now, evaluate the given expression:
$\tan\left(\frac{f(x)+g(x)}{6}\right) = \tan\left(\frac{\pi + \frac{\pi}{2}}{6}\right)$

$= \tan\left(\frac{3\pi}{12}\right) = \tan\left(\frac{\pi}{4}\right) = 1$

Q10. Consider the function $f:\mathbb{R}\to[0,1]$ defined by $f(x)=2\sin^4\left(\frac{\pi x}{4}\right)+2\cos^4\left(\frac{\pi x}{4}\right)-1$

If T is the fundamental period of $f(x)$, then which of the following statements is **CORRECT** ?

(A) $f(x)$ is an into function and $T=4$ (B) $f(x)$ is an onto function and $T=2$

(C) $f(x)$ is an onto function and $T=4$ (D) $f(x)$ is an into function and $T=2$

Ans. B

Sol. Using the identity $\sin^4\theta+\cos^4\theta=1-2\sin^2\theta\cos^2\theta=1-\frac{1}{2}\sin^2(2\theta)$, we simplify $f(x)$ as follows:

$f(x)=2\left(1-\frac{1}{2}\sin^2\left(\frac{\pi x}{2}\right)\right)-1$

$f(x)=2-\sin^2\left(\frac{\pi x}{2}\right)-1=1-\sin^2\left(\frac{\pi x}{2}\right)=\cos^2\left(\frac{\pi x}{2}\right)$

Expressing $f(x)$ using the double-angle formula:

$f(x)=\dfrac{1+\cos(\pi x)}{2}$

1. Range and Onto/Into Property:

Since $-1\le\cos(\pi x)\le 1$, we get:

$0\le\dfrac{1+\cos(\pi x)}{2}\le 1\implies$ Range of $f=[0,1]$

Given Codomain $=[0,1]$.

Since Range $=$ Codomain, $f(x)$ is an onto function.

2. Fundamental Period:

The fundamental period of $\cos(\pi x)$ is $T=\dfrac{2\pi}{\pi}=2$.

Thus, the fundamental period of $f(x)$ is $T=2$.

Hence, $f(x)$ is an onto function and $T=2$.

Correct Option: (B)

Q11. Let $U=\{x\in\mathbb{Z}:-3\le x\le 5\}$ be the universal set. Consider the sets $A=\{x\in U:x^2-2x-3\le 0\}$,$B=\{x\in U:x \text{ is a prime number}\}$

If $A\Delta B$ denotes the symmetric difference of sets A and B, then the number of elements in $A\Delta B$, denoted by $n(A\Delta B)$, is equal to

(A) 3 (B) 4 (C) 5 (D) 6

Ans. B

Sol. The universal set is given by: $U=\{-3,-2,-1,0,1,2,3,4,5\}$

For set A: $x^2-2x-3\le 0\implies(x-3)(x+1)\le 0\implies -1\le x\le 3$

Since $x\in\mathbb{Z}$, we have: $A=\{-1,0,1,2,3\}$

For set B: The prime numbers contained in U are $2,3$, and 5. Thus: $B=\{2,3,5\}$

Now, the symmetric difference $A\Delta B$ is given by: $A\Delta B=(A\setminus B)\cup(B\setminus A)$

Here, $A\setminus B=\{-1,0,1\}$; $B\setminus A=\{5\}$

Therefore, $A\Delta B = \{-1, 0, 1, 5\}$

Hence, the number of elements is $n(A\Delta B) = 4$.

Q12. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by
$f(x) = \max\{|x^2 - 2x|,\ k(x-1)+1\}$, where $k \in \mathbb{R}$. Let S be the set of all values of k for which the number of points at which $f(x)$ is not differentiable is minimum. Then S is equal to:

(A) {-1, 1} (B) $\{1\}$ (C) $\{0\}$ (D) R

Ans. C

Sol. Observe that at $x = 1$: $g(1) = |1 - 2| = 1$ and $h(1) = k(0) + 1 = 1$

So, the straight line $y = h(x)$ always passes through the point $(1, 1)$ for any $k \in \mathbb{R}$.

For $x \in (0, 2)$, $g(x) = 2x - x^2$, which has its local maximum at $x = 1$ with $g'(1) = 0$.

Case 1: When $k = 0$ $h(x) = 1$.

The function becomes $f(x) = \max\{|x^2 - 2x|, 1\}$.

Solving $|x^2 - 2x| = 1$: $x^2 - 2x - 1 = 0 \implies x = 1 \pm \sqrt{2}$.

Since $|x^2 - 2x| \le 1$ for $x \in [1 - \sqrt{2}, 1 + \sqrt{2}]$, we have:
$f(x) = \begin{cases} x^2 - 2x, & x \le 1 - \sqrt{2} \\ 1, & 1 - \sqrt{2} < x < 1 + \sqrt{2} \\ x^2 - 2x, & x \ge 1 + \sqrt{2} \end{cases}$

At $x = 1 - \sqrt{2}$: $f'((1-\sqrt{2})^-) = 2(1 - \sqrt{2}) - 2 = -2\sqrt{2}$

$f'((1-\sqrt{2})^+) = 0$

Thus, $f(x)$ is non-differentiable at $x = 1 - \sqrt{2}$.

Similarly, at $x = 1 + \sqrt{2}$: $f'((1+\sqrt{2})^-) = 0$, $f'((1+\sqrt{2})^+) = 2(1 + \sqrt{2}) - 2 = 2\sqrt{2}$

Thus, $f(x)$ is non-differentiable at $x = 1 + \sqrt{2}$.

For all other points, $f(x)$ is smooth.

Hence, for $k = 0$, $f(x)$ is non-differentiable at **exactly 2 points**.

Case 2: When $k \neq 0$ Near $x = 1$, $g(x) = 2x - x^2$ with $g'(1) = 0$, while $h'(1) = k \neq 0$.

Thus, $h(x)$ and $g(x)$ intersect transversally at $x = 1$, making $x = 1$ a point of non-differentiability.

In addition, $h(x)$ intersects $g(x) = 2x - x^2$ at $x = 1 - k$, and intersects $g(x) = x^2 - 2x$ at two outer points $x = \dfrac{k + 2 \pm \sqrt{k^2 + 8}}{2}$.

This gives at least 4 points of non-differentiability for any $k \neq 0$.

Therefore, the minimum number of non-differentiable points is 2, which occurs only when $k = 0$.

Q13. Consider the function $f : R \to R$ defined by $f(x) = \lim\limits_{n \to \infty} \dfrac{x^{2n+1} + x^2 + 1}{x^{2n} + 1}$

Let $A = \{x \in \mathbb{R} : f(x) \geq 2\}$ and $B = \{x \in \mathbb{R} : f(x) \leq x + 1\}$. Then the set $B \setminus A$ is equal to:

(A) $(-\infty, -1) \cup [0, 2)$ (B) $(-\infty, -1] \cup (0, 2]$ (C) $(-\infty, -1) \cup (0, 2]$ (D) $(-\infty, -1] \cup [0, 2)$

Ans. A

Sol. First, evaluate $f(x)$ for different intervals of x:

1. For $|x| < 1$:

As n $\to \infty$, $x^{2n} \to 0$ and $x^{2n+1} \to 0$.

$$f(x) = \frac{0 + x^2 + 1}{0 + 1} = x^2 + 1$$

2.For $|x| > 1$:

Divide numerator and denominator by x^{2n}:

$$f(x) = \lim_{n\to} \frac{x + x^{2-2n} + x^{-2n}}{1 + x^{-2n}} = x$$

3. For $x = 1$: $f(1) = \frac{1+1+1}{1+1} = \frac{3}{2}$

4. For $x = -1$: $f(-1) = \frac{-1+1+1}{1+1} = \frac{1}{2}$

Now, determine Set A = {x ∈ R : f(x) ≥ 2}:

For |x| < 1: $x^2 + 1 \leq 2 \quad \Rightarrow x^2 \leq 1$, no solution in (–1, 1).

For |x| > 1 : $x \leq 2 \Rightarrow x \in [2, \infty)$.

For x = ± 1: f(1) = 3/2 < 2 and f(-1) = 1/2 < 2.

Hence, A = [2, ∞).

Next, determine Set $B = \{x \in: f(x) \leq x + 1\}$:

For |x| > 1: $x \geq x + 1 \Rightarrow 0 \geq 1$, which is always true for (–∞, -1) ∪ (1, ∞).

For |x| < 1: $x^2 + 1 \geq x + 1 \Rightarrow x(x - 1) \geq 0 \quad \Rightarrow x \in [0, 1)$.

For x = 1: f(1) = 3/2 ≥ 1 + 1 = 2 (True, so $1 \in$ B).

For x = –1: f(-1) = 1/2 ≥ –1 + 1 = 0 (False, so $-1 \notin$ B).

Hence, B = (–∞, –1) ∪ [0, ∞).

Finally, calculate B\A :

$B \setminus A = ((-\infty, -1) \cup [0, \infty)) \setminus [2, \infty) = (-\infty, -1) \cup [0, 2)$

$BA = ((-\infty, -1)[0,))[2,) = (-\infty, -1)[0, 2)$

Q14. Let $[t]$ denote the greatest integer less than or equal to t, and $\{t\}$ denote the fractional part of t. Evaluate the following limit:

$$\lim_{x\to 0^+} \frac{3x\left[\frac{2}{x}\right] + 4\left[\frac{\sin x}{x}\right] + \left\{\frac{x}{\sin x}\right\}}{2\left\{\frac{\sin x}{x}\right\} + 3\left[\frac{x}{\tan x}\right] + 5x^2\left[\frac{1}{x^2}\right]}$$

(A) 1 (B) 2 (C) $\frac{3}{5}$ (D) $\frac{6}{7}$

Ans. D

Sol. To evaluate the limit as $x \to 0^+$, we analyze each term individually using the properties of standard inequalities, greatest integer function, and fractional part function for $x \in \left(0, \frac{\pi}{2}\right)$:

1. Evaluation of $3x\left[\frac{2}{x}\right]$: Using $u-1<[u]\le u$, we get: $3x\left(\frac{2}{x}-1\right)<3x\left[\frac{2}{x}\right]\le 3x\left(\frac{2}{x}\right)$

$\implies 6-3x<3x\left[\frac{2}{x}\right]\le 6$

By the Squeeze Theorem, $\lim\limits_{x\to 0^+} 3x\left[\frac{2}{x}\right]=6$.

2. Evaluation of $\left[\frac{\sin x}{x}\right]$: For $x\in\left(0,\frac{\pi}{2}\right)$, $\sin x<x \implies 0<\frac{\sin x}{x}<1$.

Therefore, $\left[\frac{\sin x}{x}\right]=0$, so $4\left[\frac{\sin x}{x}\right]\to 0$.

3. Evaluation of $\left\{\frac{x}{\sin x}\right\}$:

For $x\in\left(0,\frac{\pi}{2}\right)$, $\frac{x}{\sin x}>1$ and $\lim\limits_{x\to 0^+}\frac{x}{\sin x}=1^+$.

Thus, for small $x>0$, $1<\frac{x}{\sin x}<2 \implies \left\{\frac{x}{\sin x}\right\}=\frac{x}{\sin x}-1$.

Hence, $\lim\limits_{x\to 0^+}\left(\frac{x}{\sin x}-1\right)=1-1=0$.

4. Evaluation of $\left\{\frac{\sin x}{x}\right\}$:

Since $0<\frac{\sin x}{x}<1$, we have $\left\{\frac{\sin x}{x}\right\}=\frac{\sin x}{x}$.

Hence, $\lim\limits_{x\to 0^+} 2\left\{\frac{\sin x}{x}\right\}=2\times 1=2$.

5. Evaluation of $\left[\frac{x}{\tan x}\right]$:

For $x\in\left(0,\frac{\pi}{2}\right)$, $\tan x>x \implies 0<\frac{x}{\tan x}<1$.

Therefore, $\left[\frac{x}{\tan x}\right]=0$, so $3\left[\frac{x}{\tan x}\right]\to 0$.

6. Evaluation of $5x^2\left[\frac{1}{x^2}\right]$:

Using the Squeeze Theorem similarly: $5-5x^2<5x^2\left[\frac{1}{x^2}\right]\le 5 \implies \lim\limits_{x\to 0^+} 5x^2\left[\frac{1}{x^2}\right]=5$

Substituting all these individual limits back into the original expression:

Limit $=\frac{6+0+0}{2+0+5}=\frac{6}{7}$

Q15. Let $f:\left[0,\frac{2}{\pi}\right]\to\mathbb{R}$ be a function defined by

$f(x)=\begin{cases} x^{\alpha}\sin\left(\frac{1}{x}\right)+2x-1, & x\in\left(0,\frac{2}{\pi}\right] \\ \beta, & x=0 \end{cases}$ If $f(x)$ is continuous on $\left[0,\frac{2}{\pi}\right]$ and differentiable at $x=0$, then which of the following options is **CORRECT** ?

(A) $\beta=-1,\alpha>1$, and $f(x)=0$ has at least one root in $\left(0,\frac{2}{\pi}\right)$

(B) $\beta=-1,\alpha>0$, and $f(x)=0$ has no root in $\left(0,\frac{2}{\pi}\right)$

(C) $\beta = 0, \alpha > 1$, and $f(x) = 0$ has at least one root in $\left(0, \frac{2}{\pi}\right)$

(D) $\beta = -1, \alpha < 1$, and $f(x) = 0$ has no root in $\left(0, \frac{2}{\pi}\right)$

Ans. A

Sol. **1. For continuity at $x = 0$:**

We require $\lim\limits_{x \to 0^+} f(x) = f(0) = \beta$.

$\lim\limits_{x \to 0^+} \left(x^\alpha \sin\left(\frac{1}{x}\right) + 2x - 1\right) = 0 + 0 - 1 = -1$ (Since $\left|\sin\left(\frac{1}{x}\right)\right| \leq 1$, for $\alpha > 0$, $\lim\limits_{x \to 0^+} x^\alpha \sin\left(\frac{1}{x}\right) = 0$ by Squeeze Theorem).

Thus, $\beta = -1$.

2. For differentiability at $x = 0$:

The right-hand derivative at $x = 0$ is:

$$f'(0^+) = \lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^+} \frac{h^\alpha \sin\left(\frac{1}{h}\right) + 2h - 1 - (-1)}{h}$$

$$= \lim_{h \to 0^+} \left(h^{\alpha - 1} \sin\left(\frac{1}{h}\right) + 2\right)$$

For this limit to exist finitely, we must have $\alpha - 1 > 0 \implies \alpha > 1$.

3. Applying Intermediate Value Theorem:

Since $\alpha > 1$, $f(x)$ is continuous on the closed interval $\left[0, \frac{2}{\pi}\right]$.

$f(0) = -1 < 0$

$$f\left(\frac{2}{\pi}\right) = \left(\frac{2}{\pi}\right)^\alpha \sin\left(\frac{\pi}{2}\right) + 2\left(\frac{2}{\pi}\right) - 1$$

$$= \left(\frac{2}{\pi}\right)^\alpha + \frac{4}{\pi} - 1$$

Since $\left(\frac{2}{\pi}\right)^\alpha > 0$ and $\frac{4}{\pi} > 1$, we get $f\left(\frac{2}{\pi}\right) > 0$.

Since $f(0) < 0$ and $f\left(\frac{2}{\pi}\right) > 0$, by the Intermediate Value Theorem, there exists at least one root $c \in \left(0, \frac{2}{\pi}\right)$ such that $f(c) = 0$.

Hence, the correct option is **(A)**.

Q16. Let $A = \{1, 2\}$ and $B = \{a, b\}$. If $C = A \times B$, then the total number of reflexive relations that can be defined on the set C is equal to :

(A) 2^8 (B) 2^{10} (C) 2^{12} (D) 2^{16}

Ans. C

Sol. The number of elements in set A is $n(A) = 2$ and in set B is $n(B) = 2$.

The set $C = A \times B$ is the Cartesian product of sets A and B, so the number of elements in C is given by:

$n(C) = n(A) \times n(B) = 2 \times 2 = 4$

For any finite set S containing m elements, the total number of reflexive relations defined on S is given by the formula 2^{m^2-m}.

Here, $m = n(C) = 4$.

Substituting $m = 4$:

Number of reflexive relations $= 2^{4^2-4} = 2^{16-4} = 2^{12}$

Q17. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} \frac{\tan x}{x} + \sin x \cdot \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 2, & x = 0 \end{cases}$$

Then at $x = 0$, the function $f(x)$ has:

(A) a removable discontinuity (B) a jump (non-removable of first kind) discontinuity

(C) an essential (oscillatory) discontinuity (D) an infinite discontinuity

Ans. A

Sol. To determine the nature of discontinuity at $x = 0$, we first evaluate $\lim_{x\to 0} f(x)$.

$$\lim_{x\to 0} f(x) = \lim_{x\to 0} \left(\frac{\tan x}{x} + \sin x \cdot \cos\left(\frac{1}{x}\right) \right)$$

Using standard limit results:

$$\lim_{x\to 0} \frac{\tan x}{x} = 1$$

For the term $\sin x \cdot \cos\left(\frac{1}{x}\right)$:

Since $-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$ for all $x \neq 0$, and $\lim_{x\to 0} \sin x = 0$, by the Sandwich Theorem:

$$\lim_{x\to 0} \left(\sin x \cdot \cos\left(\frac{1}{x}\right) \right) = 0$$

Therefore, $\lim_{x\to 0} f(x) = 1 + 0 = 1$

However, the value of the function at $x = 0$ is given as $f(0) = 2$.

Since $\lim_{x\to 0} f(x)$ exists finitely but $\lim_{x\to 0} f(x) \neq f(0)$, the function $f(x)$ has a **removable discontinuity** at $x = 0$.

Q18. Let $f(x) = \sin^4 x + \cos^4 x$ for all $x \in \mathbb{R}$, and let T be the fundamental period of $f(x)$. Consider the function $g : \left[0, \frac{T}{2}\right] \to \mathbb{R}$ defined as $g(x) = f(x)$. Which of the following statements is **CORRECT** ?

(A) $T = \pi$ and $g(x)$ is a many-one function.

(B) $T = \frac{\pi}{2}$ and $g(x)$ is a strictly increasing one-one function.

(C) $T = \frac{\pi}{2}$ and $g(x)$ is a strictly decreasing one-one function.

(D) $T = \pi$ and $g(x)$ is a strictly decreasing one-one function.

Ans. C

Sol. Simplifying the function $f(x)$: $f(x) = (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = 1 - \frac{1}{2}\sin^2 2x$

$= 1 - \frac{1}{2}\left(\frac{1 - \cos 4x}{2}\right) = \frac{3}{4} + \frac{1}{4}\cos 4x$

The fundamental period of $\cos 4x$ is $T = \frac{2\pi}{4} = \frac{\pi}{2}$.

Thus, the domain of $g(x)$ is $\left[0, \frac{T}{2}\right] = \left[0, \frac{\pi}{4}\right]$.

To check monotonicity and injectivity of $g(x)$ on $\left[0, \frac{\pi}{4}\right]$: $g'(x) = -\sin 4x$

For $x \in \left(0, \frac{\pi}{4}\right)$, $4x \in (0, \pi)$, which gives $\sin 4x > 0 \implies g'(x) < 0$.

Since $g'(x) < 0$ on $\left(0, \frac{\pi}{4}\right)$, $g(x)$ is strictly decreasing on $\left[0, \frac{\pi}{4}\right]$.

Any strictly monotonic function on an interval is one-one.

Hence, $T = \frac{\pi}{2}$ and $g(x)$ is a strictly decreasing one-one function.

Q19. Let $S = \{1, 2, 3\}$. Let a be the number of relations on S that are both reflexive and symmetric, and let b be the total number of equivalence relations on S.

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{e^{ax} - 1 - ax}{x^2}, & x > 0 \\ c, & x = 0 \\ \frac{1 - \cos(bx)}{x \sin(x)} + d, & x < 0 \end{cases}$$

If $f(x)$ is continuous at $x = 0$, then the value of $c + 2d$ is equal to:

(A) 64 (B) 71 (C) 83 (D) 56

Ans. B

Sol. First, we count the required relations on $S = \{1, 2, 3\}$ ($n = 3$):

1. A relation R on S is reflexive and symmetric if all diagonal pairs $(i, i) \in R$ for $i \in \{1, 2, 3\}$, and

for each of the $\frac{3(3-1)}{2} = 3$ non-diagonal pairs, both elements are either in R or not in R.
$a = 2^{\frac{3(3-1)}{2}} = 2^3 = 8$

2. The number of equivalence relations on S corresponds to the number of partitions of S:
- Partition of type $\{1, 2, 3\}$: 1 way
- Partition of type $\{1, 2\}, \{3\}$: 3 ways
- Partition of type $\{1\}, \{2\}, \{3\}$: 1 way
$b = 1 + 3 + 1 = 5$

Now, evaluating the limits of $f(x)$ as $x \to 0$:

Right-Hand Limit (RHL):

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \dfrac{\mathbf{e}^{8x} - 1 - 8x}{x^2}$

Using Taylor series $\mathbf{e}^{8x} = 1 + 8x + \dfrac{(8x)^2}{2!} + \ldots$:

$\lim_{x \to 0^+} f(x) = \dfrac{8^2}{2} = 32$

Left-Hand Limit (LHL):

$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \left(\dfrac{1 - \cos(5x)}{x^2} \cdot \dfrac{x}{\sin x} + d \right)$

$= \dfrac{5^2}{2} \cdot 1 + d = \dfrac{25}{2} + d$

Since $f(x)$ is continuous at $x = 0$:

$\text{RHL} = \text{LHL} = f(0)$

$c = 32$

$\dfrac{25}{2} + d = 32 \implies 2d = 64 - 25 = 39$

Thus,

$c + 2d = 32 + 39 = 71$

Q20. Let T be the fundamental period of the function $f(x) = \sin\left(\dfrac{\pi x}{4}\right) + \cos\left(\dfrac{\pi x}{3}\right)$. A relation R is defined on the set $A = \{1, 2, 3, \ldots, T\}$ by $R = \{(x, y) \in A \times A : x + 3y = T\}$
Then the number of elements in the domain of R is:

(A) 6 (B) 7 (C) 8 (D) 24

Ans. B

Sol. First, determine the fundamental period T of $f(x)$:

Period of $\sin\left(\dfrac{\pi x}{4}\right)$ is $T_1 = \dfrac{2\pi}{\pi/4} = 8$.

Period of $\cos\left(\dfrac{\pi x}{3}\right)$ is $T_2 = \dfrac{2\pi}{\pi/3} = 6$.

The fundamental period of $f(x)$ is $T = \text{LCM}(8, 6) = 24$.

Thus, $A = \{1, 2, 3, \ldots, 24\}$.

The relation R is given by $R = \{(x, y) \in A \times A : x + 3y = 24\}$.
Rearranging gives $x = 24 - 3y$.

Since $x, y \in A$:
$1 \leq 24 - 3y \leq 24 \implies 0 \leq 3y \leq 23 \implies y \in \{1, 2, 3, 4, 5, 6, 7\}$

For these values of y, the corresponding values of x are:

- $y = 1 \implies x = 21$
- $y = 2 \implies x = 18$
- $y = 3 \implies x = 15$
- $y = 4 \implies x = 12$
- $y = 5 \implies x = 9$

- $y = 6 \implies x = 6$
- $y = 7 \implies x = 3$

The domain of R is the set of first elements in the ordered pairs:
$\text{Domain}(R) = \{3, 6, 9, 12, 15, 18, 21\}$

The number of elements in the domain of R is 7.

Q1. Let $f(x) = \max\{|x-2|, x^2-4\}$ for all $x \in \mathbb{R}$.

If $L = \lim\limits_{x \to 2^-} \dfrac{f(x)-f(2)}{x-2}$ and $R = \lim\limits_{x \to 2^+} \dfrac{f(x)-f(2)}{x-2}$, and N denotes the total number of points in $\mathbb{R}$ where the composite function $g(x) = f(|x|)$ is not differentiable, then find the value of $(R-L)+N$.

Ans. 8

Sol. First, analyze $f(x) = \max\{|x-2|, x^2-4\}$.
We compare $|x-2|$ and x^2-4:
For $x \in (-3, 2)$, $x^2-4 < 2-x = |x-2|$, so $f(x) = 2-x$.
For $x \geq 2$, $x^2-4 \geq x-2 = |x-2|$, so $f(x) = x^2-4$.
For $x \leq -3$, $x^2-4 \geq 2-x = |x-2|$, so $f(x) = x^2-4$.

Hence, $f(2) = 0$.

Now evaluate the left-hand limit L:

$$L = \lim_{x \to 2^-} \frac{f(x)-f(2)}{x-2}$$
$$= \lim_{x \to 2^-} \frac{(2-x)-0}{x-2} = -1$$

Evaluate the right-hand limit R:

$$R = \lim_{x \to 2^+} \frac{f(x)-f(2)}{x-2} = \lim_{x \to 2^+} \frac{(x^2-4)-0}{x-2}$$
$$= \lim_{x \to 2^+} (x+2) = 4$$

Thus, $R - L = 4 - (-1) = 5$.

Next, consider the composite function $g(x) = f(|x|)$.
Since $|x| \geq 0$, using the definition of $f(x)$:
For $0 \leq |x| < 2 \implies x \in (-2, 2)$, $g(x) = 2 - |x|$.
For $|x| \geq 2 \implies x \in (-\infty, -2] \cup [2, \infty)$, $g(x) = |x|^2 - 4 = x^2 - 4$.

Explicitly writing $g(x)$:

$$g(x) = \begin{cases} x^2-4, & x \leq -2 \\ 2+x, & -2 < x < 0 \\ 2-x, & 0 \leq x < 2 \\ x^2-4, & x \geq 2 \end{cases}$$

Testing differentiability of $g(x)$ at $x = -2, 0, 2$:

1. At $x = -2$: LHD $= -4$, RHD $= 1 \implies$ Non-differentiable.

2. At $x = 0$: LHD $= 1$, RHD $= -1 \implies$ Non-differentiable.

3. At $x = 2$: LHD $= -1$, RHD $= 4 \implies$ Non-differentiable.

$g(x)$ is differentiable at all other points.

Therefore, $N = 3$.

Finally, $(R - L) + N = 5 + 3 = 8$.

Q2. Let $f(x)$ and $g(x)$ be defined as

$$f(x) = \begin{cases} 2 - x, & x < 1 \\ x^2 - 2x + 3, & x \geq 1 \end{cases} \text{ and } g(x) = \begin{cases} x + 3, & -4 \leq x \leq 2 \\ 7 - x, & 2 < x \leq 6 \end{cases}$$

If D is the domain of the function $h(x) = \dfrac{\sqrt{f(x) - 2}}{g(x) - 3}$, then the number of integers contained in D is equal to _____.

Ans. 9

Sol. To find the domain D of $h(x) = \dfrac{\sqrt{f(x) - 2}}{g(x) - 3}$, we must satisfy two conditions:

1. x must lie in the domain of $g(x)$, which is $[-4, 6]$.

2. The numerator requires $f(x) - 2 \geq 0 \implies f(x) \geq 2$.

- For $x < 1$: $2 - x \geq 2 \implies x \leq 0$.

- For $x \geq 1$: $x^2 - 2x + 3 \geq 2 \implies (x - 1)^2 \geq 0$, which holds for all $x \geq 1$.

Thus, $f(x) \geq 2 \implies x \in (-\infty, 0] \cup [1, \infty)$.

3. The denominator requires $g(x) - 3 \neq 0 \implies g(x) \neq 3$.

- For $-4 \leq x \leq 2$: $x + 3 \neq 3 \implies x \neq 0$.

- For $2 < x \leq 6$: $7 - x \neq 3 \implies x \neq 4$.

Thus, $x \neq 0$ and $x \neq 4$.

Taking the intersection of all conditions:

$D = [-4, 6] \cap \big((-\infty, 0] \cup [1, \infty)\big) \setminus \{0, 4\}$

$D = [-4, 0) \cup [1, 4) \cup (4, 6]$

The integer values of x in D are:

$-4, -3, -2, -1, 1, 2, 3, 5, 6$

Total number of integers = 9.

Q3. Evaluate the value of the limit:

$$L = \lim_{x \to 0^+} \frac{24\left(e^x - 1 - x - \frac{x^2}{2}\right) - 4x^3 \left\lfloor \frac{\tan x}{x} \right\rfloor + 2x^4 \left\lfloor 1 + \frac{\sin x}{x} \right\rfloor}{x^3 \{x\}}$$

where $\lfloor \cdot \rfloor$ denotes the greatest integer function and $\{\cdot\}$ denotes the fractional part function.

Ans. 3

Sol. For $x \in \left(0, \frac{\pi}{2}\right)$:

1. $\{x\} = x$

2. $\tan x > x \implies \frac{\tan x}{x} > 1 \implies \left\lfloor \frac{\tan x}{x} \right\rfloor = 1$

3. $\sin x < x \implies 0 < \frac{\sin x}{x} < 1 \implies 1 < 1 + \frac{\sin x}{x} < 2 \implies \left\lfloor 1 + \frac{\sin x}{x} \right\rfloor = 1$

Using the Maclaurin series expansion for e^x:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + O(x^5) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + O(x^5)$$

Substitute these into the numerator $N(x)$:

$$N(x) = 24\left(\frac{x^3}{6} + \frac{x^4}{24} + O(x^5)\right) - 4x^3(1) + 2x^4(1)$$

$$N(x) = \left(4x^3 + x^4 + O(x^5)\right) - 4x^3 + 2x^4 = 3x^4 + O(x^5)$$

The denominator $D(x)$ simplifies to:

$$D(x) = x^3 \cdot x = x^4$$

Now, evaluate the limit:

$$L = \lim_{x \to 0^+} \frac{3x^4 + O(x^5)}{x^4} = 3$$

Q4. Let $f : (0, 6) \to \mathbb{R}$ be a function defined by $f(x) = \tan^{-1}(x) - \tan^{-1}(\lfloor x \rfloor) - \tan^{-1}(\{x\})$ where $\lfloor \cdot \rfloor$ denotes the greatest integer function and $\{\cdot\}$ denotes the fractional part function. If N is the total number of points in the open interval $(0, 6)$ at which $f(x)$ is not differentiable, then the value of N is equal to _____.

Ans. 5

Sol. For any integer $k \in \{0, 1, 2, 3, 4, 5\}$ and $x \in (k, k+1)$, we have $\lfloor x \rfloor = k$ and $\{x\} = x - k$.

Thus, $f(x) = \tan^{-1}(x) - \tan^{-1}(k) - \tan^{-1}(x - k)$, which is smooth and infinitely differentiable on every open interval $(k, k+1)$.

Hence, non-differentiability can only occur at the integer points $x \in \{1, 2, 3, 4, 5\}$.

At $x = 1$:

Left-hand limit:

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (\tan^{-1}(x) - \tan^{-1}(0) - \tan^{-1}(x)) = 0$$

Right-hand limit:

$$\lim_{x \to 1^+} f(x) = \tan^{-1}(1) - \tan^{-1}(1) - \tan^{-1}(0) = 0$$

Value at $x = 1$: $f(1) = \tan^{-1}(1) - \tan^{-1}(1) - \tan^{-1}(0) = 0$.

Since LHL = RHL = $f(1)$, $f(x)$ is **continuous** at $x = 1$.

Now checking derivatives at $x = 1$:

Left-hand derivative: $f'(1^-) = \frac{d}{dx}(0) = 0$.

Right-hand derivative:

$$\lim_{x \to 1^+} f'(x) = \lim_{x \to 1^+} \left(\frac{1}{1 + x^2} - \frac{1}{1 + (x-1)^2}\right) = \frac{1}{2} - 1 = -\frac{1}{2}$$

Since $f'(1^-) \neq f'(1^+)$, $f(x)$ is **not differentiable** at $x = 1$.

At $x = k \in \{2, 3, 4, 5\}$:

Left-hand limit:

$\lim\limits_{x \to k^-} f(x) = \tan^{-1}(k) - \tan^{-1}(k-1) - \tan^{-1}(1)$

$= \tan^{-1}\left(\dfrac{1}{1 + k(k-1)}\right) - \dfrac{\pi}{4}$

Since $k(k-1) > 0$ for $k \geq 2$, $\dfrac{1}{1 + k(k-1)} < 1 \implies \tan^{-1}\left(\dfrac{1}{1 + k(k-1)}\right) \neq \dfrac{\pi}{4}$.

Thus, $\lim\limits_{x \to k^-} f(x) \neq 0$.

Right-hand limit: $\lim\limits_{x \to k^+} f(x) = 0$.

Since $\lim\limits_{x \to k^-} f(x) \neq \lim\limits_{x \to k^+} f(x)$, $f(x)$ is **discontinuous** at $x = 2, 3, 4, 5$, and therefore **not differentiable** at these 4 points.

Thus, the total number of points of non-differentiability in $(0, 6)$ is $N = 1 + 4 = 5$.

Q5. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \max\{\tan^{-1} x, -\tan^{-1} x\}$. Consider the function $g(x) = f(|x| - 1)$.

If N denotes the total number of points in $\mathbb{R}$ where $g(x)$ is non-differentiable, and $S = \sum\limits_{r=1}^{N} \tan^{-1}\left(\dfrac{1}{r^2 + r + 1}\right)$, then find the value of $100 \tan(S)$.

Ans. 60

Sol. First, simplify $f(x)$:

$f(x) = \max\{\tan^{-1} x, -\tan^{-1} x\} = |\tan^{-1} x|$

Now, substitute into $g(x)$:

$g(x) = f(|x| - 1) = |\tan^{-1}(|x| - 1)|$

To find the points of non-differentiability of $g(x)$:

1. The inner function $|x|$ has a sharp corner at $x = 0$. Since $\tan^{-1}(|0| - 1) = -\dfrac{\pi}{4} \neq 0$, the sharp corner at $x = 0$ persists.

2. The outer absolute value function $|\tan^{-1}(u)|$ creates non-differentiable points where $u = 0$, i.e., $|x| - 1 = 0 \implies x = \pm 1$.

Checking left and right derivatives:

At $x = 1$: $\lim\limits_{x \to 1^+} g'(x) = 1$ and $\lim\limits_{x \to 1^-} g'(x) = -1 \implies$ Non-differentiable.

At $x = -1$: $\lim\limits_{x \to -1^+} g'(x) = 1$ and $\lim\limits_{x \to -1^-} g'(x) = -1 \implies$ Non-differentiable.

At $x = 0$: $\lim\limits_{x \to 0^+} g'(x) = -1$ and $\lim\limits_{x \to 0^-} g'(x) = 1 \implies$ Non-differentiable.

Thus, $g(x)$ is non-differentiable at $x \in \{-1, 0, 1\}$.

Therefore, $N = 3$.

Now evaluate S:

$S = \sum\limits_{r=1}^{3} \tan^{-1}\left(\dfrac{1}{r^2 + r + 1}\right)$

Using the identity $\tan^{-1}\left(\frac{1}{r^2+r+1}\right) = \tan^{-1}(r+1) - \tan^{-1}(r)$:

$S = \sum_{r=1}^{3} \left(\tan^{-1}(r+1) - \tan^{-1}(r)\right) = \tan^{-1}(4) - \tan^{-1}(1)$

Taking tangent on both sides:

$tan(S) = \tan(\tan^{-1}(4) - \tan^{-1}(1)) = \frac{4-1}{1+4\cdot1} = \frac{3}{5}$